

## Part III

### Convex Optimization I

# Lecture Outline

① Convex Optimization

② Least Squares Problem

# Convex Optimization

In this lecture, we will study a particular nice class of optimization problems, the so-called convex optimization problems, consisting of a feasible set which is convex and a cost function which is convex on the feasible set. There are several reasons for studying this class of optimization problems:

- All minimizers are global
- First order necessary conditions are also sufficient
- Efficient algorithms exist for computing the minimizers

# Convex Optimization

The optimization problem

$$\min_{x \in R_c} f(x)$$

is called convex if the cost function  $f$  is convex on  $R_c$ . Note that this implies, by definition, that the feasible set  $R_c \subseteq \text{dom}(f)$  is convex.

## Example

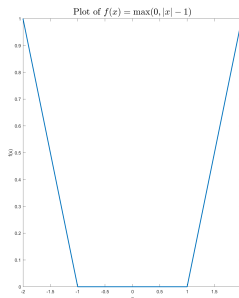
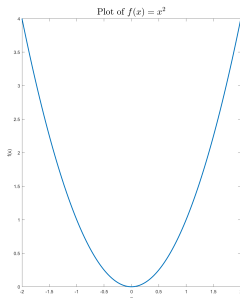
- $\min \sin(x)$  s.t.  $x \in [-\pi, 0]$  is convex, with global minimizer:  $x' = -\pi/2$
- $\min \sin(x)$  s.t.  $x \in (-\pi/2, 0]$  is convex, with no minimizer
- $\min \sin(x)$  s.t.  $x \in [-\pi, 0.5]$  is nonconvex, with global minimizer:  $x' = -\pi/2$

# Local and global minimizers

## Theorem

If  $f$  is a convex function on  $R_c$ , then

- The set of minimizers of  $f$  is convex.
- Any local minimum of  $f$  is a global minimum of  $f$ .



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### Example

Consider the convex optimization problem

$$\min_{A \in R_c} f(A)$$

with  $f(A)$  the maximum eigenvalue of the symmetric matrix  $A \in \mathbb{R}^{n \times n}$ :

$$f(A) = \max_z \{z^\top A z \mid \|z\|_2 = 1\}$$

and the feasible region  $R_c$  given by

$$R_c = \{\text{diag}(\theta_1, \dots, \theta_n) \mid 0 \leq \theta_i \leq 1, i = 1, \dots, n\},$$

or more generally by a convex subset of the vector space of  $n \times n$  symmetric matrices.

# Existence of global minimizers

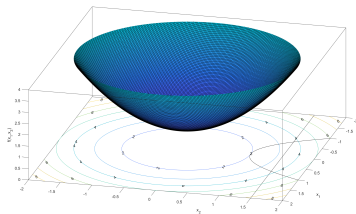
## Theorem

Let  $f \in C^1$  be a convex function on  $R_c$ . If  $x' \in R_c$  is such that

$$\nabla f(x')^T (x - x') \geq 0,$$

for all  $x \in R_c$ , then  $x'$  is a (global) minimizer of  $f$ .

Note that this says that the first order necessary conditions also are sufficient conditions.



# Maximizers of convex functions

## Theorem (Maximizers)

*Let  $f$  be a convex function on a bounded and closed (convex) set  $R_c$ . Then any maximizer lies on the boundary of  $R_c$ .*

**Technical remark:** Note that a maximizer might not exist as demonstrated by the following example: Let  $R_c = \{(r, \theta) \mid 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi\}$  denote the unit disc in  $\mathbb{R}^2$  describe in polar coordinates, and  $f(r, \theta) = 0$  for  $r < 1$  and  $f(r, \theta) = \tan(\theta/4)$ .

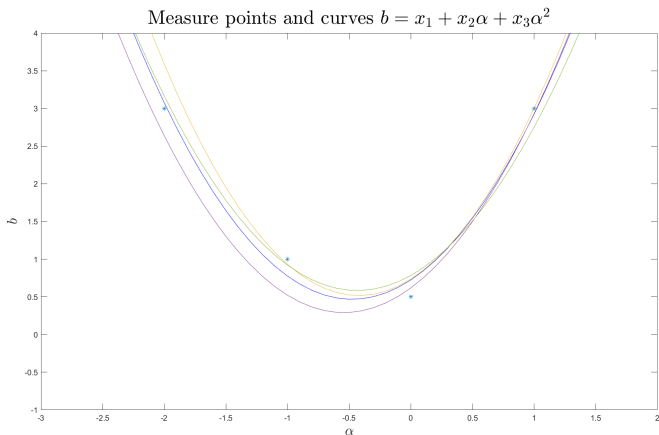


## Motivation

Given  $m$  measure points  $(\alpha_i, b_i) \in \mathbb{R}^2$ ,  $i = 1, \dots, m$ , and assume that these points approximately lie on a curve on the form

$$b = \sum_{j=1}^n x_j f_j(\alpha), \quad (\text{e.g. } f_j(\alpha) = \alpha^{j-1})$$

with  $x_i \in \mathbb{R}$  unknown parameters and  $f_j : \mathbb{R} \rightarrow \mathbb{R}$  known functions.



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Hence we may write

$$b_i \approx \sum_{j=1}^n x_j f_j(\alpha_i), \text{ for } i = 1, \dots, m,$$

or in matrix notation

$$b \approx Ax, \quad A = [a_{ij} = f_j(\alpha_i)] \in \mathbb{R}^{m \times n}.$$

The error we make is  $\|r\|$  where  $r = b - Ax$ , called the residual. Note that

$$b = Ax + r.$$

## Least Squares Problem

We assume

- $m \geq n$  (choose enough data points)
- $\|\cdot\| = \|\cdot\|_2$  (we want to minimize length).

Least squares problem (linear regression):

For given (design) matrix  $A$  and (observation) vector  $b$ , we want to determine a (parameter) vector  $x$  such that the squared error  $\|r\|_2^2 = \|b - Ax\|_2^2$  is minimized.

That is, the least squares problem can be formulated as the unconstrained optimization problem:

$$\min_x \|b - Ax\|_2^2$$

The term “least squares” come from the fact that we are trying to minimize the sum of squares

$$\|b - Ax\|_2^2 = \sum (b_i - (Ax)_i)^2.$$

Sometimes a least squares solution is written as  $\hat{x}$ . That is,

$$\hat{x} \in \arg \min_x \|b - Ax\|_2^2$$

## Solution

The least squares problem:

$$\min_x \|b - Ax\|_2^2$$

Is convex since the cost function

$$f(x) = \|b - Ax\|_2^2 = x^\top A^\top Ax - 2b^\top Ax + b^\top b$$

is quadratic and the Hessian

$$H_f(x) = A^\top A$$

is positive semi definite in general and positive definite if  $\text{null}(A) = \{0\}$  (which happens iff  $A$  has full rank since we assume  $m \geq n$ ).

Hence, the solutions to the least squares problem is given as the solutions to

$$0 = \nabla f(x) = 2A^\top Ax - 2A^\top b$$

or equivalent as the solutions to the normal equation

$$A^\top Ax = A^\top b$$

## Solution

The solutions to the least squares problem:

$$\min_x \|b - Ax\|_2^2$$

is given as the solutions to the normal equation

$$A^\top Ax = A^\top b$$

which in the case of  $\text{null}(A) = \{0\}$  simplifies to the solution

$$\hat{x} = (A^\top A)^{-1} A^\top b = A^\dagger b$$

with  $A^\dagger = (A^\top A)^{-1} A^\top$  the pseudoinverse of  $A$ .

Note that in general there are infinitely many solutions to the least squares problem. However, this case does not occur very often in applications.

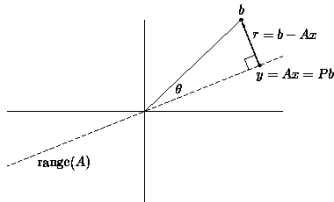
## A geometric approach

The solution to the least squares problem can be obtained geometrically since

$$Pb = A\hat{x} \quad (1)$$

with  $P = A(A^T A)^{-1} A^T$  the orthogonal projection onto  $\text{Range}(A)$ . Hence by multiplying (1) with  $A$  from the left we obtain the normal equation

$$A^T A x = A^T b$$



**Technical remark:** If  $\text{null}(A) \neq \{0\}$  the matrix  $A$  above needs to be replaced by another matrix whose columns constitute a basis for  $\text{Range}(A)$ .

## Example

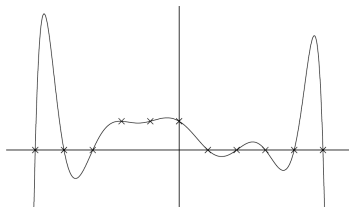


Figure 11.1. *Degree 10 polynomial interpolant to eleven data points. The axis scales are not given, as these have no effect on the picture.*

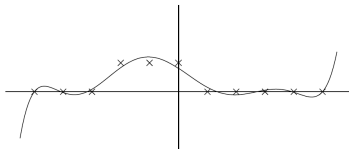


Figure 11.2. *Degree 7 polynomial least squares fit to the same eleven data points.*

## Least norm problem

If  $A$  does not have full rank, the normal equation

$$A^*Ax = A^*b$$

has infinite many solutions. In that case, one may include the least squares problem in another (larger) optimization problem in order to obtain uniqueness.

### Example (Least norm problem)

$$\min_{A^*Ax=A^*b} \|x\|$$

E.g., if  $A = \text{diag}(1, 1, 0)$ ,  $\|\cdot\| = \|\cdot\|_2$  and  $b = (1, 0, 0)$ , then  $x = (1, 0, 0)$  is a least norm solution.